

IS 733 - Lab 11

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Part I: Distributional Hypothesis

Please use the Distributional Hypothesis Concept to find the solution, show your work step by step

Example 1

1. A piece of _____ is on the plate.
2. Everyone enjoys eating _____.
3. You can cut _____ with a knife.
4. We make _____ from milk.

Potential Answers: "cheese," "cake," or "butter"

SOLUTION:

- Sentence 1: "A piece of _____ is on the plate." - **We could fit all the 3, but "cheese" and "cake" are more likely to be served as a piece.**
- Sentence 2: "Everyone enjoys eating _____." - **All could be used as the answer.**
- Sentence 3: "You can cut _____ with a knife." - **All three could be cut with the knife.**
- Sentence 4: "We make _____ from milk." - **Only "cheese" and "butter" are made from milk; "cake" is not.**

Distribution Table

	CHEESE	CAKE	BUTTER
STATEMENT 1	1	1	0
STATEMENT 2	1	1	1
STATEMENT 3	1	1	1
STATEMENT 4	1	0	1

Thus, based on the observations the best fit would be **"Cheese"** as it fits all contexts, especially being made from milk and commonly served as a piece.

Example 2

1. The _____ is parked in the driveway.
2. He bought a new _____ for his birthday.
3. _____ can drive really fast.
4. People often wash their _____ on the weekends.

Potential Answers: "car," "truck," or "motorcycle"

SOLUTION:

- Sentence 1: " The ____ is parked in the driveway. " - **All can be parked in a driveway.**
- Sentence 2: " He bought a new _____ for his birthday." - **All can be bought for a birthday.**
- Sentence 3: " _____ can drive really fast." – **Comparitively, "Car" and "motorcycle" can drive fast than a "truck".**
- Sentence 4: " People often wash their _____ on the weekends.” - **People commonly wash their "car" or "truck," less commonly a "motorcycle."**

Distribution Table

	CAR	TRUCK	MOTORCYCLE
STATEMENT 1	1	1	1
STATEMENT 2	1	1	1
STATEMENT 3	1	0	1
STATEMENT 4	1	1	0

Thus, based on the observations the best fit would be **“Car”** as it is the most common in all contexts, especially for washing and being parked in a driveway.

Example 3

1. I read an interesting _____ last night.
2. Many people enjoy a good _____ before bed.
3. _____ often has chapters and a cover.
4. You can borrow a _____ from the library.

Potential Answers: "book," "novel," or "story" – these terms appear in contexts related to reading and library usage.

SOLUTION:

- Sentence 1: “I read an interesting _____ last night.” - **All can be read and be interesting.**
- Sentence 2: “Many people enjoy a good _____ before bed.” - **All can be enjoyed**
- Sentence 3: “ _____ often has chapters and a cover.” - **Only "book" and "novel" have chapters and a cover; "story" does not.**
- Sentence 4: “You can borrow a _____ from the library.” - All can be borrowed, but "book" is the most general.

Distribution Table

	BOOK	NOVEL	STORY
STATEMENT 1	1	1	1
STATEMENT 2	1	1	1
STATEMENT 3	1	1	0
STATEMENT 4	1	0	0

Thus, based on the observations the best fit would be "**Book**" as it fits all contexts, especially having chapters and a cover.

Part II: Practice the NLP model to classify data stories

Link to dataset

Show = level 1; Tell = Level 2 and 3

You will be asked to replicate the results from this paper (Figure 6 zero shot results, cross-validation and leave one-plot out); feature extraction steps are as follows:

For each sentence

- **we converted it into lowercase**
- **removed punctuation**
- **The sentences were then split into tokens, and stop words were removed**

- *Tokenization and lemmatization*
- *feature extraction was performed using TF-IDF vectorization*

You will apply ML classification models such as logistic regression, Support Vector Machine (SVM), or Naive Bayes (NB). Optionally, you are also welcome to try other classifiers not used in the paper, such as Random Forest (RF) to see whether it outperformed the existing results.

Bonus: Try additional representations/embedding methods, and compare with the current result

Version	Experiment	Model	Training Data			
			Combined	Zero Shot	One Shot	Two Shot
LLM Generated Stories	Cross Validation	Logistic Regression	0.93	0.85	0.79	0.86
		Naive Bayes	0.92	0.79	0.78	0.82
		SVM	0.94	0.82	0.79	0.83
	Leave One Plot Out	Logistic Regression	0.95	0.94	0.92	0.95
		Naive Bayes	0.95	0.94	0.93	0.95
		SVM	0.95	0.94	0.94	0.95
Student Generated Stories	Train/Validate	Logistic Regression	0.77	0.68	0.69	0.78
		Naive Bayes	0.77	0.67	0.70	0.77
		SVM	0.76	0.67	0.68	0.77

Figure 6: ML model performance (Area Under Curve) in discriminating whether a given narrative sentence is “show” versus “tell”

SOLUTION:

The results show that traditional models like SVM and Logistic Regression can achieve good performance in classifying narrative levels when TF-IDF is used. Leave-One-Plot-Out validation gives a better measure of generalization across unseen stories. The variation in results suggests that further optimization and experimentation are needed to improve model robustness and handle data complexity effectively. Due to the lack of detailed information about the experimental environment, reproducing the exact results shown in the original analysis was not possible, but the findings here align with the overall trends observed in similar studies.

GPT USAGE STATEMENT:

I utilized AI assistance for the following aspects of this task:

- **Code Generation:** Generating boilerplate code for data loading (pandas), text preprocessing (NLTK, string), TF-IDF vectorization (sklearn), model instantiation (sklearn), and evaluation loops (train/test split, LeaveOneGroupOut CV).
 - **Debugging:** Identifying and resolving errors
 - **Conceptual Clarification:** Ensuring correct implementation of cross-validation and the calculation of overall AUC across folds.
 - **Structuring the Report:** Organizing the findings and comparison sections.
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